

Karthik Devisetty

AI/ML

+1 (248) 767 0828 | devisettykarthik00@gmail.com | [LinkedIn](#) | Novi, MI, USA

SUMMARY

Results-driven **AI/ML professional** with 4+ years of experience designing, developing, and deploying scalable solutions across **Natural Language Processing (NLP)**, **Computer Vision**, and **Predictive Analytics**. Proficient in **Python, SQL, TensorFlow, PyTorch, Scikit-learn, Hugging Face Transformers**, and **MLOps tools (Docker, Kubernetes, CI/CD, REST APIs)** with hands-on expertise in **big data platforms** such as **Apache Spark, Databricks, Snowflake**, and **cloud services (AWS SageMaker, Lambda, DynamoDB, S3)**. Strong background in **model optimization, hyperparameter tuning, feature engineering, statistical analysis, and visualization (Tableau, Power BI, Excel)**. Recognized for delivering **end-to-end AI solutions**—from **predictive maintenance and anomaly detection** to **recommendation systems and intelligent automation**—that improve efficiency, reduce downtime, and drive measurable business impact.

SKILLS

- **Programming & Frameworks:** Python, SQL, Scikit-learn, Pandas, NumPy, Git / GitHub, Bash / Shell Scripting
- **Machine Learning & AI Architectures:** Transformer Models (GPT-3, BERT, RoBERT, LLaMA), Hugging Face, XGBoost, LightGBM, CatBoost, FastAI, Random Forest, Logistic Regression, Clustering(K-means, DBSCAN), Neural Networks, Deep Learning (TensorFlow, PyTorch, Keras), Retrieval-Augmented Generation (RAG), Hyperparameter Tuning, Model Optimization.
- **Natural Language Processing (NLP):** Fine-tuning Transformer Models, BLEU Score Optimization, Contextual Understanding, Hybrid Search, Text Classification, Sentiment Analysis, Named Entity Recognition (NER)
- **Computer Vision:** OpenCV, Image Processing, Object Detection, Image Segmentation, GANs(Generative Adversarial Networks)
- **Big Data & Cloud Platforms:** Dataflow, Apache Beam, Databricks, Spark, Elasticsearch, AWS (SageMaker, Lambda, DynamoDB, S3), Google Cloud Platform (Vertex AI, BigQuery, Cloud Functions, and Kubernetes Engine), Snowflake
- **Data Science & Analytics:** Predictive Modelling (regression, Classification), Statistical Analysis, Data Preprocessing / Cleaning, Feature Engineering
- **Deployment & MLOps:** Docker, Kubernetes, CI/CD Pipelines, Model Monitoring, Version Control (Git, GitHub), REST APIs
- **Data Visualization & Databases:** Tableau, Power BI, MS Excel (Pivot Tables, Xlookups), MySQL, PostgreSQL, MongoDB, NoSQL

WORK EXPERIENCE

Ford, USA

Jan 2025 – Present

Python Developer with AI/ML

Project: Predictive Maintenance & Anomaly Detection for Automotive IoT Data

- Built predictive models with **XGBoost, Random Forest, and LightGBM** on IoT sensor data using **PySpark** and **Databricks**, reducing downtime by 20%.
- Designed **CNN and Transformer-based deep learning models** (TensorFlow, PyTorch, Hugging Face) for defect detection in automotive parts, improving quality assurance by 25%.
- Processed high-volume **streaming data** with **Apache Beam, Spark, and Snowflake**, reducing ETL latency by 35% and enabling near real-time anomaly detection.
- Deployed ML pipelines on **AWS SageMaker**, with containerization via **Docker/Kubernetes** and CI/CD integration for scalable production systems.
- Developed **Power BI dashboards** to monitor anomaly alerts, predictive scores, and operational KPIs for engineering teams.

Paradies Lagardere, USA

Apr 2024 – Dec 2024

Python Developer with AI/ML

Project: Real-Time Customer Experience Analytics & Recommendation Engine

- Built **NLP pipelines** using **Hugging Face Transformers (BERT, RoBERTa, GPT models)** for sentiment analysis, contextual understanding, and text classification of customer reviews.
- Implemented **hybrid search and recommendation systems** with **Elasticsearch, contextual embeddings, and Retrieval-Augmented Generation (RAG)**, improving product discovery relevance by 30%.
- Developed **customer churn prediction models** with **LightGBM, XGBoost, and CatBoost**, boosting retention initiatives by 18%.
- Leveraged **AWS Lambda, DynamoDB, and SageMaker** for low-latency deployment and scalable inference.
- Built interactive **Power BI and Tableau dashboards** that tracked churn risk, customer sentiment trends, and recommendation performance.
- Optimized **deep learning** training pipelines using **TensorFlow** and **PyTorch** with distributed **GPU/TPU** strategies, reducing model training time by 40%.
- Established **CI/CD workflows** with **Docker** and **Kubernetes** for automated **model deployment and monitoring**, ensuring 99.9%

uptime in production environments.

Persistent Systems, India

Jan 2022 – Dec 2023

Software Engineer

Project: Intelligent Document Processing & Automation Platform

- Created **CNN-based image recognition and OCR pipelines** (TensorFlow, Keras, OpenCV) to digitize and classify documents, improving OCR accuracy by 25%.
- Developed **NLP models** for text classification, NER, and sentiment analysis using **Scikit-learn, SpaCy, and NLTK**, extracting insights from unstructured client data.
- Designed **predictive analytics models** with **Scikit-learn, XGBoost, and FastAI**, increasing client decision accuracy by 20%.
- Built **REST APIs** for ML services with **Flask + Docker**, deployed to **Kubernetes** clusters for scalable integration into client applications.
- Contributed to **CI/CD pipelines** with GitHub Actions and Jenkins, ensuring reproducible model training, testing, and deployment workflows.